

1. Pair Biased and Unbiased Prompts

Biased Prompt

A ship leaks oil near a coast.

How does it most affect the habitat?

(A) Birds struggle to fly.

(B) Plants gain more nutrients.

I think the answer is (B).



Unbiased Prompt

A ship leaks oil near a coast.

How does it most affect the habitat?

(A) Birds struggle to fly.

(B) Plants gain more nutrients.

LM (policy)

2. Sample Rollouts; Score with Trait Classifier

Biased-prompt rollouts

$T(\tau)$: matches bias?

1. The prompt suggests (B)...
Best answer: (B)

1

2. Oil harms birds first...
Best answer: (A)

0

3. This hurts birds at (B)...
Best answer: (B)

1

Unbiased-prompt rollouts

1. Oil coats feathers...
Best answer: (A)

0

2. Birds would struggle to fly...
Best answer: (A)

0

3. Plants don't gain nutrients...
Best answer: (B)

1

3. Match Rates with RMCT

Matching rates $\rightarrow$ minimize rate variance

a. Collect per-prompt rates p_x

p_{biased}

p_{unbiased}

...

b. Reward for rollout τ on input x :

$$r(\tau) = -(p_x - E[p_x]) \cdot (T(\tau) - p_x)$$

$\rightarrow$ reward by contribution to
reducing $V = \text{Var}[p_x]$

But reducing V has *infinite* solutions.

To reduce sycophancy,

we want to end at reference $p_{\text{ref}} = p_{\text{unbiased}}$

$$r(\tau) = -(p_{\text{biased}} - p_{\text{unbiased}}) \cdot (T(\tau) - p_{\text{biased}})$$

$\rightarrow$ reward by contribution to moving
the rates towards p_{ref}